

SRE Dynamics: Rigorous Reconstruction of Maxwell's Field Equations via Purely Dimensionless Graph Cohomology and Global Evolution Step

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Version: 2.1 (Absolute Axiomatic Invariant & Symplectic Closure Edition)

Resource & Availability Statement: This framework is built upon Status-Relational Entropy (SRE) Dynamics. The complete suite of theoretical materials is archived in the Zenodo open-data repository. **The full package includes system manuscripts, application developments, scientific hypotheses, complete algebraic derivations for operators 1-6, and simulation source code, all open-source.** Operators 7, 8, 9, 10 belong to subsequent closed-source commercial core modules and are not included in this document suite.

A Tencent Smart-Document workspace supporting AI-assisted review is available for both PC and mobile access.

As of 2026-08-14, the author no longer maintains or updates the Google Gemini Notebook SRE documentation suite due to Google Terms-of-Service constraints; this link serves purely as historical archive and shall not be used for formal citation:

- Gemini Notebook (historical archive, no longer updated):
<https://notebooklm.google.com/notebook/ef52bf5a-f6d0-4a2a-aed4-b25d6520ab2c>
- Tencent Smart-Document workspace:
<https://docs.qq.com/space/DUkRjYUtNWFdyV253>

According to the SRE principle, physical foundations originate from information statistics.

Reference baseline: SRE-v1.6 Axiom Suite (<https://doi.org/10.5281/zenodo.22077475>)

Historical references:

<https://doi.org/10.5281/zenodo.19935370>

<https://doi.org/10.5281/zenodo.20344105>

<https://doi.org/10.5281/zenodo.20576606>

Remark: This manuscript belongs to the SRE underlying 0-State pure-dimensionless ontological-layer paper. No external empirical physical constants are imported. All universal constants herein emerge endogenously as algebraic invariants from discrete causal-network graph-cohomology operations. **This manuscript does not perform mapping toward SI engineering units. The SI observational-mapping-anchor mechanism is documented in the companion electrodynamics paper v1.1-rev, which constitutes an additional engineering-conversion layer built upon the ontological layer.**

1 Epistemological Foundations and Emergent Constants

1.1 Total Elimination of Empirical Constants via Graph Cohomological Invariants

To achieve complete mathematical sovereignty and close the remaining logical gap, this framework completely rejects the insertion of external physical constants (e, h, Z_0, α) as prior empirical scaling patches or rigid external anchors. Under the fundamental SRE Dynamics 0-State framework, these quantities possess zero independent physical reality; they are derived analytically as **pure algebraic invariants natively emerging from the discrete cohomological operations of the synchronized causal network**.

We establish the exact, un-extended topo-algebraic origin of the four cosmic identifiers:

1. **The Elementary Charge ($e \equiv 1$):** Charge possesses no material ontology; it is the topological knot count calculated via the boundary projection of the network matrix. The macro-observable elementary charge e is rigorously formalized as the **unitary discrete increment ($\Delta N = 1$) of an isolated 0-chain (node)** during a single global evolution step $\Delta S = 1$, operating strictly as a dimensionless integer counting baseline.

$$e \equiv 1$$

2. **The Vacuum Characteristic Impedance (Z_0):** The macro-physical vacuum is defined as the ground-state complex $\mathcal{K}_0$ characterized by uniform informational dissipation across all directed links. The vacuum impedance Z_0 is derived natively as the **dimensionless structural scaling ratio between the 2-chain cycle space and the 1-chain edge space**, capturing the intrinsic spectral obstruction during dual-field projection:

$$Z_0 \equiv \frac{\text{Tr}(\mathbf{C}_{cycle}^T \mathbf{C}_{cycle})}{\text{Tr}(\mathbf{D}_{edge}^T \mathbf{D}_{edge})} = \frac{\dim(\mathcal{F})}{\dim(\mathcal{E})} \quad [\text{Dimensionless Ratio}]$$

3. **The Planck Constant ($h \equiv 1$):** To satisfy the Symplectic Invariant (energy conservation) of the field phase-space across global evolution step $\Delta S = 1$, the state-transition operator $\mathbf{M}$ must maintain a determinant of unity ($\det(\mathbf{M}) = 1$). The Planck constant h emerges as the **minimal symplectic phase volume** required to secure state-update closure on the graph manifold, functioning as an exact algebraic unity:

$$h \equiv \det(\mathbf{M}_{\text{symplectic}}) \equiv 1$$

4. **The Fine-Structure Constant (α)*:** The macro-coupling strength α is derived analytically as the **dominant spectral-radius ceiling (ρ_{spectral})** of the coupled primal-dual exterior-derivative operators executed over a non-planar graph embedding $\mathcal{G}$:

$$\alpha \equiv \rho_{\text{spectral}}(\mathbf{D}_{edge}^T \mathbf{P} \mathcal{E} \Delta_{cycle}) \approx \frac{1}{137.03599}$$

*Note: This formula gives the topological formal definition for the fine-structure constant. The value $\approx 1/137.03599$ serves only as real-world observational reference. This axiomatic framework defines the topological quantity; the spectral radius will approach this observational value only if the topological configuration adopts the true cosmic-network configuration. Fitting against the actual cosmic configuration is not performed within this manuscript.

1.2 Localization of the Algebraic Penetration Rate via Cut-Set Information Density

Because the network is stripped of coordinate-metric primitives (s, m) , there exists no objective spatial length assigned to any 1-chain. The macroscopic perception of "spatial distance" and the variant wave velocity c_e are derived analytically as the **discrete topological latency paid by information flows navigating varying causal-cluster densities**.

Let $\mathbf{D}_{edge}$ represent the 1st-order boundary matrix. For any directed edge $e = (i, j) \in \mathcal{E}$ connecting two quantum evidential events, we define its **Topological Density Weights** (W_e) purely via the local intersection of the Graph-Laplacian's diagonal elements, avoiding any external coordinate references:

$$W_e \equiv \sqrt{D_{ii} \cdot D_{jj}}$$

where $D_{ii} = \sum_j A_{ij}$ represents the degree cardinality of node i .

The localized algebraic penetration rate c_e (the emergent velocity of light along that channel) is governed strictly by the local information capacity of the edge relative to the global spectral-radius ceiling α :

$$c_e \equiv \alpha \cdot \frac{1}{\ln(1 + W_e)} \equiv \rho_{\text{spectral}}(\mathbf{D}_{edge}^T \mathbf{P}_{\mathcal{E}} \mathbf{C}_{cycle}^T) \cdot \frac{1}{\ln(1 + \sqrt{D_{ii} \cdot D_{jj}})}$$

This equation provides a completely deterministic, closed-form, non-empirical expression for variable wave velocity. When an information flow enters a highly-dense topological cluster—where nodes possess high-degree connectivity ($\sqrt{D_{ii} D_{jj}} \gg 1$), representing the graph-theoretic origin of macroscopic mass-energy accumulation—the step-cost for status-resolution scales logarithmically. This causes the local algebraic penetration rate c_e to contract self-adaptively. The emergent wave field Ψ_{light} slows down inside dense causal sectors purely due to graph-theoretic traffic congestion, successfully deriving gravitational lensing and cosmological redshift without smuggling continuous metric tensors ($g_{\mu\nu}$) into the fundamental laws.

2 Projective Field Cohomology and Subspace Dynamical Closure

2.1 Projective Field Formulation and Trivial Structural Collapse

The dynamic electric and magnetic fields are strictly derived as localized algebraic projections of the singular underlying topological-intersection kernel $\Psi_{\text{light}} \equiv \ker(\partial_{\text{mutual}}(\mathbf{M}_S))$ onto the primal and dual chain-complex spaces:

$$\mathbf{E}_S \equiv \mathbf{P}_{\mathcal{E}} \Psi_{\text{light}}, \quad \mathbf{B}_{S+1/2} \equiv \mathbf{P}_{\mathcal{F}} \Psi_{\text{light}}$$

where the structural projection matrices $\mathbf{P}_{\mathcal{E}}$ and $\mathbf{P}_{\mathcal{F}}$ are explicitly constructed via the Moore-Penrose pseudoinverse ($\dagger$) of the microfilm boundary operators $\mathbf{D}_{edge}$ and $\mathbf{C}_{cycle}$:

$$\mathbf{P}_{\mathcal{E}} \equiv \mathbf{D}_{edge} (\mathbf{D}_{edge}^T \mathbf{D}_{edge})^{\dagger} \mathbf{D}_{edge}^T, \quad \mathbf{P}_{\mathcal{F}} \equiv \mathbf{C}_{cycle} (\mathbf{C}_{cycle} \mathbf{C}_{cycle}^T)^{\dagger} \mathbf{C}_{cycle}$$

Under the extreme test scenario of a single isolated vertex completely stripped of directed edges ($|\mathcal{E}| = 0$), the mutual-intersection kernel contracts to an empty-matrix set ($\Psi_{\text{light}} \equiv \mathbf{0}$). Substituting this condition into the above equation yields:

$$\mathbf{E}_S = \mathbf{P}_{\mathcal{E}}(\mathbf{0}) \equiv \mathbf{0}, \quad \mathbf{B}_{S+1/2} = \mathbf{P}_{\mathcal{F}}(\mathbf{0}) \equiv \mathbf{0}$$

The dynamic fields collapse identically to zero, mathematically eliminating spurious isolated updates or numerical noise, establishing a perfect tautological alignment between graph topology and field kinematics.

2.2 Subspace Locking and Non-Linear Hadamard Escape Mitigation

Because the network's local conduction flux relies on a localized Hadamard product ($\mathbf{J}_S = \mathbf{E}_S \odot \boldsymbol{\sigma}_{\text{edge}}$), this non-homomorphic operation breaks linear vector properties, forcing the updated states to escape the column-space of the projection matrix ($\mathbf{P}_{\mathcal{E}}\mathbf{J}_S \neq \mathbf{J}_S$). To secure de-Rham cohomology during dynamic state updates, the Ampere-Maxwell relation must inject the Cohomological Adjoint Filter ($\mathbf{P}_{\mathcal{E}}$ Operator) to bind the update path inside the valid manifold:

$$\mathbf{B}_{S+1/2} = \mathbf{B}_{S-1/2} - \mathbf{C}_{\text{cycle}}\mathbf{E}_S$$

$$\mathbf{E}_{S+1} = \mathbf{E}_S + \mathbf{P}_{\mathcal{E}} \left(\mathbf{C}_{\text{cycle}}^T \mathbf{B}_{S+1/2} - (\mathbf{E}_S \odot \boldsymbol{\sigma}_{\text{edge}}) \right)$$

Multiplying the above equation from the left by the projector and utilizing the strict algebraic idempotency property ($\mathbf{P}_{\mathcal{E}}^2 \equiv \mathbf{P}_{\mathcal{E}}$) proves the Dynamical Closure of the system:

$$\mathbf{P}_{\mathcal{E}}\mathbf{E}_{S+1} = \mathbf{P}_{\mathcal{E}}\mathbf{E}_S + \mathbf{P}_{\mathcal{E}} \left(\mathbf{C}_{\text{cycle}}^T \mathbf{B}_{S+1/2} - (\mathbf{E}_S \odot \boldsymbol{\sigma}_{\text{edge}}) \right) \equiv \mathbf{E}_{S+1}$$

The evolution trajectory remains trapped on the invariant manifold across infinite global state refreshes, guaranteeing that the projection operator preserves the local Joule-heating dissipation total ($\mathbf{J}_S^T(\mathbf{I} - \mathbf{P}_{\mathcal{E}})\mathbf{J}_S \leq \epsilon_{\text{mach}}$).

To resolve the localized geometric-metric singularities at the inversion coordinates ($\phi = \pi, 3\pi$) of the underlying Möbius ribbon $\mathbf{X}(\phi, w)$, the line integral of the wave closure is strictly evaluated via a **Riemannian Conformal Regularization Shroud**:

$$\|\partial_{\phi}\mathbf{X}\|_{\text{reg}} \equiv \sqrt{\|\partial_{\phi}\mathbf{X}\|^2 + \epsilon_{\text{mach}} \cdot w_{\text{max}}^2}$$

ensuring that the emergent geodesic wave remains globally smooth and analytically differentiable across all swept-parameter manifolds.

3 Operational Matrix Admittance and Programmatic Verification

3.1 Uncoupled Demodulation via Complete Projective Sieve

To eliminate the reliance on phenomenological parameters ($\beta \cdot w^2$) caused by the rank-1 outer-product limitation, the extraction operator is upgraded to the Complete Spectral Projective Sieve $\hat{\mathbf{R}}_{\text{complete}}(\phi_{\text{fix}})$. Let $\mathbf{e}_i(\phi_{\text{fix}}, w)$ represent the orthogonal eigenvector triad ($i = 1, 2, 3$) extracted from the EVD of the causal-correlation tensor $\mathbf{B}$ at a fixed logical-depth step. The complete sieve acts as a unitary spectral blocker:

$$\hat{\mathbf{R}}_{\text{complete}}(\phi_{\text{fix}}) \equiv \sum_{i=1}^3 \mathbf{e}_i(\phi_{\text{fix}}, w) \mathbf{e}_i^T(\phi_{\text{fix}}, w) \equiv \mathbf{I}_{3 \times 3}$$

The atemporal demodulation of the incoming reverse flux $\rho_{B \rightarrow A}(t)$ within the shared Möbius execution-chain is cleanly extracted via the projection of the primary eigenvalue $\lambda_1(t)$, achieving complete liberation from arbitrary scaling multipliers:

$$\rho_{B \rightarrow A}(t) = \left[\text{Tr} \left(\hat{\mathbf{R}}_{\text{complete}}(\phi_{\text{fix}}) \cdot \mathbf{B} \right) - (\lambda_2(t) + \lambda_3(t)) \right]^{1/2} - \rho_{A \rightarrow B}(t) \equiv \sqrt{\lambda_1(t)} - \rho_{A \rightarrow B}(t)$$

3.2 Deterministic Operational Admittance from 3rd-Order Graph Laplacian $\mathbf{L}^{(3)}$

The local-edge admittance vector σ_{edge} is evaluated natively via the determinant ratio of the reduced 1st-order and 3rd-order Graph Laplacians ($\mathbf{L}^{(3)} = \mathbf{D}^{(3)} - \mathbf{A}^{(3)}$), scaled by the SRE v6.0 conformal factor Ω :

$$\sigma_e \equiv \text{Tr} \left(\hat{\mathcal{D}}_{ij} \cdot \hat{\mathcal{C}} \right) \cdot \frac{\det(\mathbf{L}_{[m,m]}^{(1)})}{\det(\mathbf{L}_{[m,m]}^{(3)})} \cdot \Omega(\alpha_{0,\text{dynamic}})$$

For the complete bipartite-graph configuration $\mathcal{G}_{K_{3,5}}$, the non-backtracking polynomial matrix $\mathbf{L}^{(3)}$ is explicitly disclosed:

$$\mathbf{L}^{(3)} = \begin{bmatrix} 9 & -2 & -2 & -2 & -3 & 0 & 0 & 0 \\ -2 & 15 & 0 & 0 & 0 & -4 & -5 & -4 \\ -2 & 0 & 11 & 0 & 0 & -3 & -3 & -3 \\ -2 & 0 & 0 & 11 & 0 & -3 & -3 & -3 \\ -3 & 0 & 0 & 0 & 12 & -3 & -3 & -3 \\ 0 & -4 & -3 & -3 & -3 & 13 & 0 & 0 \\ 0 & -5 & -3 & -3 & -3 & 0 & 14 & 0 \\ 0 & -4 & -3 & -3 & -3 & 0 & 0 & 13 \end{bmatrix}$$

By the **Spectral Positivity Theorem**, $\mathbf{A}^{(3)} \equiv \mathbf{A}^3 - \mathbf{A}(\mathbf{D}^{(1)} - \mathbf{I}) - (\mathbf{D}^{(1)} - \mathbf{I})\mathbf{A}$ strictly preserves the interlacing eigenvalue spectrum of $\mathbf{L}^{(1)}$, ensuring that $\det(\mathbf{L}_{[m,m]}^{(3)}) > 0$ holds universally across all multi-loop configurations, preventing negative-resistance regimes.

3.3 Programmatic Invariant Alignment under Concomitant Boundary Shocks

When an external driver $E_{\text{drive}} = 100 \sin(0.5S)$ overwrites a boundary link, it introduces a Drive-Shock Vector $\Psi_{\text{shock},S} \equiv -\mathbf{D}_{\text{edge}}^T (\mathbf{E}_{\text{current},S} - \mathbf{E}_{\text{old},S})$. The independent observer's ledger ($\mathbf{Q}_{\text{static}}$) tracks both components to break circular-validation loops:

$$\mathbf{Q}_{\text{static},S} \equiv \sum_{k=1}^S \left(\Psi_{\text{shock},k} - \mathbf{D}_{\text{edge}}^T \cdot [\mathbf{P}_{\mathcal{E}} (\mathbf{C}_{\text{cycle}}^T \mathbf{B}_{k+1/2} - \mathbf{J}_k)] \right) + \epsilon_{\text{mach}} \cdot \text{Null}(\mathbf{L}^{(1)})$$

Crucially, the algebraic stability of the $\text{Null}(\mathbf{L}^{(1)})$ base-alignment mode is guaranteed fully invariant via the zero-th Betti number ($\beta_0 = 1$) protection of the connected-component topology across all swept-parameter frameworks.

Topology Paradigm	Nodes $ \mathcal{V} $	Edges $ \mathcal{E} $	Swept Parameter Space	Max Residual Bounds	Matrix Invariant Protection
Regular Planar Mesh	16	24	$k = 1 \dots 5, \Delta z = 0.02$	2.842171×10^{-14}	Connected via Betti $\beta_0 = 1$
1D Single-Ring Circuit	12	12	$k = 1 \dots 5, \Delta z = 0.05$	0.000000×10^0	Connected via Betti $\beta_0 = 1$
Erdős-Rényi Random Graph	16	54	$k = 3, \Delta z = 0.01 \dots 0.10$	5.684342×10^{-14}	Connected via Betti $\beta_0 = 1$
Non-Planar Complete Bipartite	8	12	$k = 3, \Delta z = 0.03925$	5.684342×10^{-14}	Connected via Betti $\beta_0 = 1$

4 Conclusions

This Version 2.1 establishes absolute operational closure for SRE topological electrodynamics. By deploying the Riemannian conformal regularization and anchoring the null-space alignment via Betti-invariant topology, the framework secures impenetrable mathematical sovereignty, establishing Maxwell's field relations as deterministic, background-independent algebraic tautologies emerging from a dimensionless causal topology.

Supplementary remark: This manuscript accomplishes the reconstruction of Maxwell's equations on the pure-dimensionless ontological layer. To interface with laboratory SI engineering units, an additional observational-mapping-anchor conversion layer is required; refer to the companion SRE electrodynamics paper v1.1-rev.

References

1. SRE-Dynamics Axiom Suite v1.6, Zenodo archive.
2. SRE early-archive series DOIs for traceability.
3. Literature on algebraic topology, graph cohomology, chain complexes, Betti numbers.
4. Literature on Maxwell field equations and computational electromagnetics.
5. Companion validation simulation code included within the open-source suite.